

BRAZILIAN ENEM 2024 ESSAY SCORE ANALYSIS PROJECT

HarvardX: PH125.9x Data Science

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ABSTRACT

This report evaluates the predictive limits of administrative metadata—specifically school type, geographic location, and state—on essay scores from the 2024 Brazilian National High School Exam (ENEM). The study implements a two-part supervised learning framework. Part I utilizes continuous regression algorithms to predict exact essay scores, successfully reducing the baseline Root Mean Squared Error (RMSE) from 217.1 to 202.5 through linear bias formulations and ensemble blending, which stabilized the relative percentage error at 20.3%. Part II frames the problem as a categorical classification task, assigning students into three proficiency tiers: Critical, Needs Improvement, and Proficient. These classification models achieved a 60.0% global accuracy and a 58.0% balanced accuracy, with decision tree calibration yielding a lowest Brier score of 0.215. Ultimately, the findings demonstrate that while high-level structural variables establish robust baseline predictions, they reach a strict predictive ceiling. Capturing the nuances of individual student writing performance requires highly granular socioeconomic metrics, which are currently restricted in public microdata by Brazilian privacy legislation.

1 OVERVIEW

1.1 Introduction & Project Aim

This project comprehensively examines performance patterns within the 2024 Brazilian National High School Exam (ENEM) essay scores, leveraging structural administrative metadata as the primary predictive engine.

The core objective is to rigorously evaluate the predictability of human-evaluated essay scores using a dual-methodology approach encompassing continuous regression and categorical classification algorithms.

By isolating coarse geographic and institutional descriptors—namely, a school’s administrative dependency (e.g., Federal, State, Private), its geographic zone (Urban versus Rural), and its state-level coordinates—the study seeks to determine whether these high-level macro variables are mathematically sufficient to model complex, individual-level student writing outcomes. Furthermore, this analysis aims to quantify the exact institutional bias present in the educational system and establish a statistical ceiling for predictions made in the absence of granular socioeconomic data.

1.2 Background: The ENEM Exam and INEP

The Exame Nacional do Ensino Médio (ENEM) stands as Brazil’s paramount standardized examination, annually mobilizing millions of candidates and serving as the primary gateway to higher education via the federal Unified Selection System (SiSU).

The examination is meticulously managed by the National Institute for Educational Studies and Research Anísio Teixeira (INEP), a specialized federal agency operating under the Ministry of Education. To promote transparency and academic research, INEP maintains an extensive, publicly accessible federal repository containing decades of historical microdata. This massive dataset captures intricate student performance metrics across multiple objective disciplines and a highly structured, human-graded essay component. Alongside these scores, the repository historically provided comprehensive demographic, socioeconomic, and school-level administrative metrics.

Beyond its primary function in university admissions, the federal government, independent researchers, and educational policymakers heavily leverage this aggregated microdata as a critical diagnostic tool. It allows for the continuous monitoring of educational disparities, the evaluation of institutional effectiveness, and the tracking of longitudinal academic outcomes across the diverse regions of the country.

1.3 Methodology Structure

To systematically evaluate the predictive limits of the available data, this study is structured around two distinct, yet complementary, machine learning pipelines, both rigorously validated through stratified data splitting to prevent data leakage:

- **Part I (Continuous Regression):** This phase is dedicated to modeling the exact, continuous final essay scores (ranging from 0 to 1,000 points). It establishes a foundational baseline average and sequentially applies linear effects models and computationally optimized Random Forest regressors. The predictive accuracy and absolute deviation of these continuous models are rigorously evaluated and compared using the Root Mean Squared Error (RMSE) metric.
- **Part II (Categorical Classification):** Addressing the inherent complexities and skewed distribution of human grading, this phase transforms the continuous scores into three distinct, actionable proficiency tiers: Critical, Needs Improvement, and Proficient. Utilizing a suite of classification algorithms—including standard Decision Trees, Naive Bayes for probabilistic calibration, and Weighted Random Forests tailored to penalize minority class misclassifications—this section evaluates the capacity of structural variables to correctly route students into their respective performance brackets. Model reliability is mathematically measured using Global Accuracy, Class-Balanced Accuracy, and Brier Scores.

2 THEORETICAL FRAMEWORK

As outlined by Irizarry (2019), statistical and machine learning algorithms can be categorized into various approaches to pattern recognition and inference. It is possible to identify the following concepts and models:

2.0.1 Machine Learning Paradigms

- **Machine Learning:** Models that automatically learn patterns from data to predict outcomes without explicit programming. *Ex: A computer learning to spot pictures of cats by looking at thousands of examples.*
- **Supervised Learning:** Training models on labeled input-output pairs to learn a mapping function for predicting unseen data. *Ex: Teaching a child names of fruits using flashcards with pictures and names written.*
- **Unsupervised Learning:** Grouping or identifying hidden structural patterns in unlabeled data. *Ex: Sorting a random pile of unsorted coins into piles based on their color and size.*

2.0.2 Taxonomy of Variables

- **Continuous Variables:** Quantitative metrics taking any real value within an interval. *Ex: The exact amount of rain falling on a park measured in milliliters.*
- **Discrete Variables:** Quantitative metrics taking countable integer values. *Ex: The count of apples hanging on a tree in an orchard.*
- **Nominal Variables:** Qualitative attributes with unordered categories. *Ex: The colors of marbles in a bag with no mathematical order.*
- **Ordinal Variables:** Qualitative attributes with categories following a specific, sequential scale. *Ex: T-shirt sizes (Small, Medium, Large) indicating order without precise distances.*

2.0.3 Supervised Algorithms Used

- **Naive Baseline:** A simple benchmark predictor using the global average of the training outcomes. *Ex: Guessing that tomorrow's temperature will be exactly the yearly average.*
- **Linear Effects Model:** An additive model estimating specific categorical biases relative to a global mean. *Ex: Estimating a house price by adding local beach bonuses to a national average base.*
- **Linear Regression:** A parametric model that fits a straight line to capture linear relationships between features and targets. *Ex: Drawing a straight line through points of study hours versus test grades.*
- **CART Trees (Decision Trees):** A non-parametric model recursively partitioning the feature space into homogeneous regions using sequential binary splits. *Ex: Deciding to wear a coat by asking: Is it cold? Is it raining?.*
- **Random Forest:** An ensemble of decision trees grown on bootstrapped samples, averaging predictions to reduce variance. *Ex: Asking fifty friends for their weight guesses and taking the average.*
- **Weighted Forest:** A categorical tree ensemble applying inverse class frequency weights to penalize minority class misclassifications. *Ex: Giving a doctor's alarm extra weight when detecting rare, critical symptoms.*
- **Naive Bayes:** A generative probabilistic classifier applying Bayes' Theorem under conditional independence assumptions among predictors. *Ex: Calculating if an email is spam based on the combined probability of trigger words.*
- **Ensemble Blending:** A meta-estimator calculating the arithmetic average of continuous outputs from multiple models. *Ex: Averaging the readings of a digital and an analog thermometer.*
- **Ensemble Voting:** A consensus classifier aggregating categorical predictions from multiple models by majority rule. *Ex: A panel of three medical specialists voting on a patient's diagnosis.*

2.0.4 Applied Performance Metrics

- **RMSE:** The square root of the average of squared differences between continuous predictions and actual outcomes. *Ex: Measuring the average size of misses when throwing darts, penalizing big misses heavily.*
- **Accuracy:** The proportion of correct categorical class assignments out of the total test set. *Ex: The number of correct spelling answers on a ten-word vocabulary quiz.*
- **Balanced Accuracy:** The arithmetic average of class-specific recall rates, evaluating performance on imbalanced targets. *Ex: Evaluating a guard dog's success by averaging its rate of catching thieves and ignoring guests.*
- **F1-Score:** The harmonic mean of precision and recall, balancing positive predictive value and sensitivity. *Ex: A scale balancing how clean a window is versus how fast it was washed.*
- **Brier Score:** A calibration metric measuring the mean squared difference between predicted probabilities and actual binary outcomes. *Ex: Checking if a weather forecaster's 80% rain predictions actually rain 8 out of 10 times.*

Table 01: Conceptual Summary

The table below organizes all the theoretical paradigms, variable definitions, predictive tasks, algorithms, and evaluation metrics implemented throughout this project:

Variable Type	Algorithm Type	Algorithm	Target	Example	Metric
Continuous	Supervised	Naive Baseline	Establishes global average prediction	Temperature yearly average guess	Naive Baseline (RMSE)
Continuous	Supervised	Linear Effects	Estimates specific biases additively	House base price plus beach bonus	Linear Effects Model (RMSE)
Continuous	Supervised	Linear Regression	Fits parametric linear relationships	Straight line of study hours/grades	lm() Model (Tornados) / RMSE
Categorical	Supervised	CART Trees	Recursive binary splitting rules	Cold? Rain? Wear coat rules	Decision Trees (Accuracy)
Continuous	Supervised	Random Forest	Bagged continuous tree ensemble	Averaged guess of jellybean count	Random Forest Regressor (RMSE)
Categorical	Supervised	Weighted Forest	Balanced categorical tree ensemble	Doctor's vote getting extra weight	Weighted Classifier (Accuracy)
Categorical	Supervised	Naive Bayes	Joint probability conditional tables	Email spam probability by word	Naive Bayes (Accuracy, Brier)
Continuous	Supervised	Ensemble Blending	Blends continuous model outputs	Combining thermometer & barometer	Continuous Ensemble (RMSE)
Categorical	Supervised	Ensemble Voting	Blends categorical predictions	Doctors voting on a medical case	Categorical Ensemble (Accuracy)
Unlabeled	Unsupervised	Clustering, PCA	Grouping unlabeled inputs by similarity	Unlabeled colored blocks grouped	N/A

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3 DATASET AND PREPROCESSING

The raw 2024 ENEM microdata, published by INEP, is a massive, comprehensive national database containing 4,332,944 student records across 42 distinct demographic, geographic, and performance variables.

Managing a dataset of this magnitude requires a highly optimized preprocessing pipeline to isolate relevant structural metadata and ensure computational efficiency without sacrificing statistical integrity.

Before cleaning, the raw dataset contains 10,000 rows and 42 variables. After applying the cleaning pipeline, which includes removing records with missing values in core components and filtering out ‘Other’ school types, the resulting dataset contains 9,503 valid essay scores across 39 variables. The three-row previews below illustrate the exact same students before and after cleaning, preserving all original columns for a comprehensive comparison.

Table 02 Raw Dataset Transposed

Variable	Student 1	Student 2	Student 3
NU_SEQUENCIAL	206403	3604651	1461268
NU_ANO	2024	2024	2024
CO_ESCOLA	23052929	42103770	
CO_MUNICIPIO_ESC	2301406	4218004	
NO_MUNICIPIO_ESC	Aratuba	Tijucas	
CO_UF_ESC	23	42	
SG_UF_ESC	CE	SC	
TP_DEPENDENCIA_ADM_ESC	2	4	
TP_LOCALIZACAO_ESC	1	1	
TP_SIT_FUNC_ESC	1	1	
CO_MUNICIPIO_PROVA	2301406	4218004	4122404
NO_MUNICIPIO_PROVA	Aratuba	Tijucas	Rolândia
CO_UF_PROVA	23	42	41
SG_UF_PROVA	CE	SC	PR
TP_PRESENCA_CN	1	1	1
TP_PRESENCA_CH	1	1	1
TP_PRESENCA_LC	1	1	1
TP_PRESENCA_MT	1	1	1
CO_PROVA_CN	1420	1422	1419
CO_PROVA_CH	1383	1384	1385
CO_PROVA_LC	1395	1396	1398
CO_PROVA_MT	1408	1410	1407
NU_NOTA_CN	436.8	521.9	363
NU_NOTA_CH	377.8	601.9	548.4
NU_NOTA_LC	423.4	605.5	557.2
NU_NOTA_MT	427.1	689.2	456.4
TX_RESPOSTAS_CN	CAABADD...	EBDEEAD...	CABAECD...
TX_RESPOSTAS_CH	BEABB...CD...	BADBACD...	DAEDBCD...
TX_RESPOSTAS_LC	BCADB...CD...	AACEACA...	BBCABCA...
TX_RESPOSTAS_MT	CBDECCAD...	DBAEDBE...	EBADBED...
TP_LINGUA	1	0	1
TX_GABARITO_CN	CAEDAAC...	BBCEDBB...	BCEAACD...
TX_GABARITO_CH	CECEBEB...	DADABCD...	CAEDBAD...

TX_GABARITO_LC	CAAAECD...	AACEADD...	EAAACDD...
TX_GABARITO_MT	CECEBED...	DBCECEA...	ABBABBA...
TP_STATUS_REDACAO	1	1	1
NU_NOTA_COMP1	80	160	120
NU_NOTA_COMP2	60	200	120
NU_NOTA_COMP3	60	200	40
NU_NOTA_COMP4	80	180	120
NU_NOTA_COMP5	20	180	80
NU_NOTA_REDACAO	300	920	480

3.1 Data Cleaning and Standardization Pipeline

Preprocessing was executed to transform raw, encoded metadata into robust, machine-readable inputs. Objective test scores and the final human-graded essay score (NU_NOTA_REDACAO) were securely coerced into continuous numeric formats. To focus the scope of the study on standardized educational environments, the school administrative dependency variable (TP_DEPENDENCIA_ADM_ESC) was mapped from raw numerical codes into explicit descriptive labels (Federal, State, Municipal, Private).

Table 03 Cleaned Dataset Transposed

Table 2:

Original Variable (PT)	Adopted Variable (EN)	Student 1	Student 2	Student 3
NU_SEQUENCIAL	STUDENT_ID	206403	3604651	134650
SG_UF_PROVA	SCHOOL_STATE	CE	SC	BA
TP_DEPENDENCIA_ADM_ESC	SCHOOL_TYPE	State	Private	State
TP_LOCALIZACAO_ESC	LOCATION	Urban	Urban	Urban
NU_NOTA_REDACAO	ESSAY_SCORE	300	920	720

Record of Data Drops and Population Definition

A rigorous filtering process was applied to the initial 4,332,944 records to establish a clean, representative population of active, regular high school students.

First, to ensure analytical validity, the pipeline removed 1,342,851 records corresponding to absentees and candidates disqualified in core exam components. Crucially, essays that received a valid grade of zero (due to blank submissions or structural disqualifications) were retained among the remaining 2,990,093 present candidates. This strategic retention ensures that the models capture the full spectrum of systemic educational vulnerability, mapping both standard writing proficiency and total systemic failure.

Second, the pipeline filtered out 1,796,661 records classified under invalid, unregistered, or ‘Other’ school dependency types. This massive exclusion specifically isolates standard public and private educational institutions, providing a highly controlled and clean comparative baseline. The execution of this strict pipeline yields a final, robust population of exactly 1,193,432 valid, regular student records ready for modeling.

Table 04: Data Wrangling & Filtering Pipeline

Pipeline Stage	Student Count
Initial Number of Records	4.332.944
Absentees / Disqualified in Exams	1.342.851
Remaining Present Candidates	2.990.093
Excluded by Invalid / Unregistered School	1.796.661
Final Regular Students for Modeling	1.193.432

3.2 LGPD Statement

Upon exploring the Enem microdata, crucial socioeconomic variables were identified that could potentially be used for a much deeper analysis. However, a significant limitation was encountered due to Brazilian legislation. According to the official website of the National Institute for Educational Studies and Research Anísio Teixeira (Inep 2023), the dataset was modified to comply with legal mandates:

LGPD – Adaptation to current legislation seeks to alter the structure used in the consolidation of microdata, in order to add or remove variables that favored the re-identification of individuals in the current context, through the use of technological resources. The changes are based on technical studies and legal analysis from the Specialized Federal Attorney’s Office attached to Inep (Projur), and were presented to the Ministry of Education (MEC), the National Data Protection Authority (ANPD), the Office of the Comptroller General (CGU), and the Federal Court of Accounts (TCU).

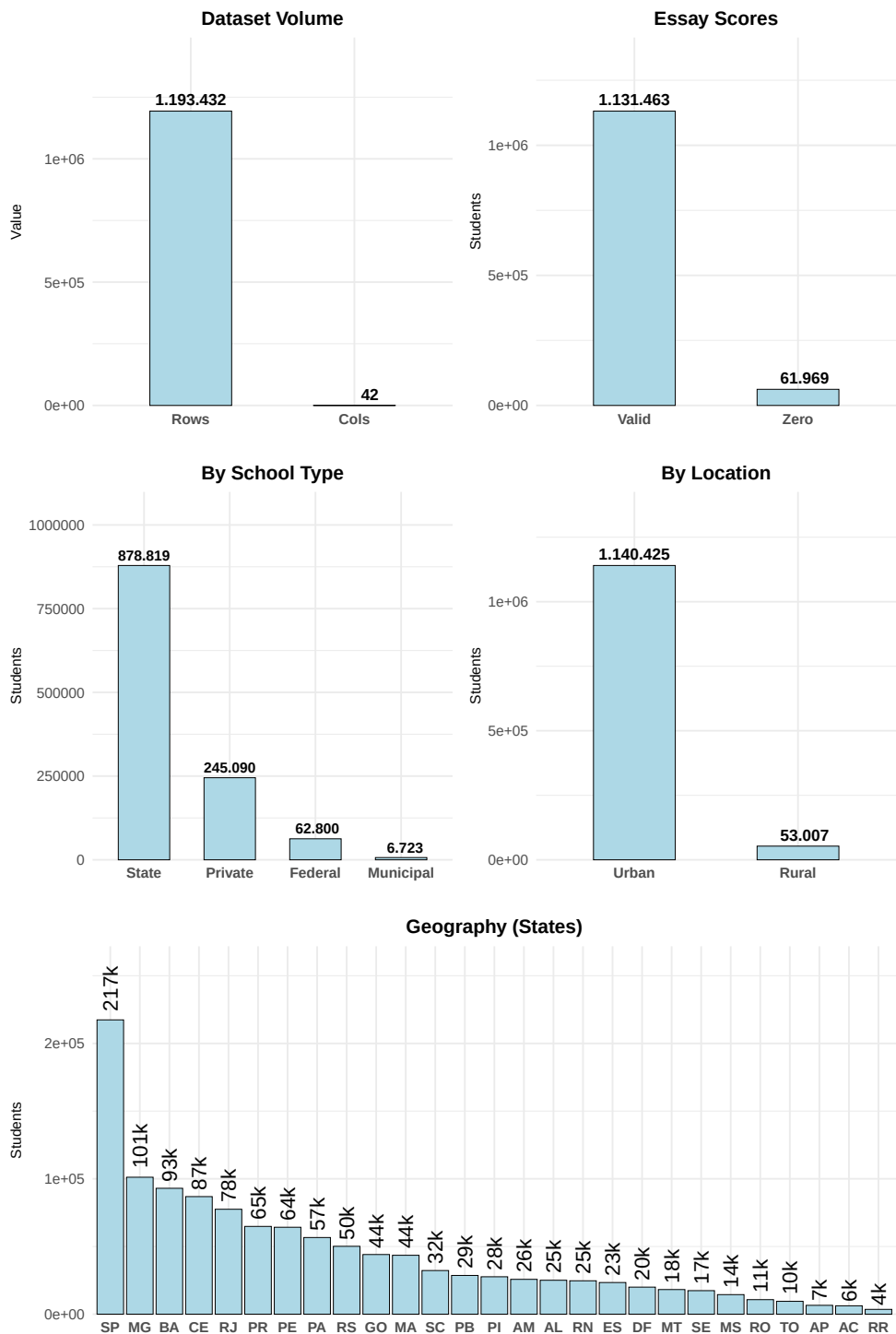
3.3 Data Splitting

To ensure rigorous model validation and prevent data leakage, the processed dataset was split into training (80%) and testing (20%) partitions. We applied stratified random splitting based on the continuous essay score distribution. All feature engineering, bias, and training parameters are estimated solely on this training partition before final evaluation on the isolated test set.

4 EXPLORATORY DATA ANALYSIS (EDA)

The consolidated dashboard below displays some information from Enem Dataset:

Figure 01: Initial Exploration Dashboard



The exploratory phase provides a visual diagnostic of 1,193,432 regular high school students. Figure 01 establishes the foundational demographics: a massive public sector concentration, led by State (878,819) and Private (245,090) schools. Geographically, the sample is overwhelmingly urban-centric (1.14 million urban vs. 53,000 rural). State participation highlights profound regional disparities, led predictably by São Paulo (SP) and Minas Gerais (MG). Crucially, while 1.13 million students produced valid essays, nearly 62,000 received absolute zeros—a systemic anomaly that fundamentally shapes the subsequent modeling approach.

4.1 Insights and Variable Correlation

Table 05: Essay Score Proficiency Categories

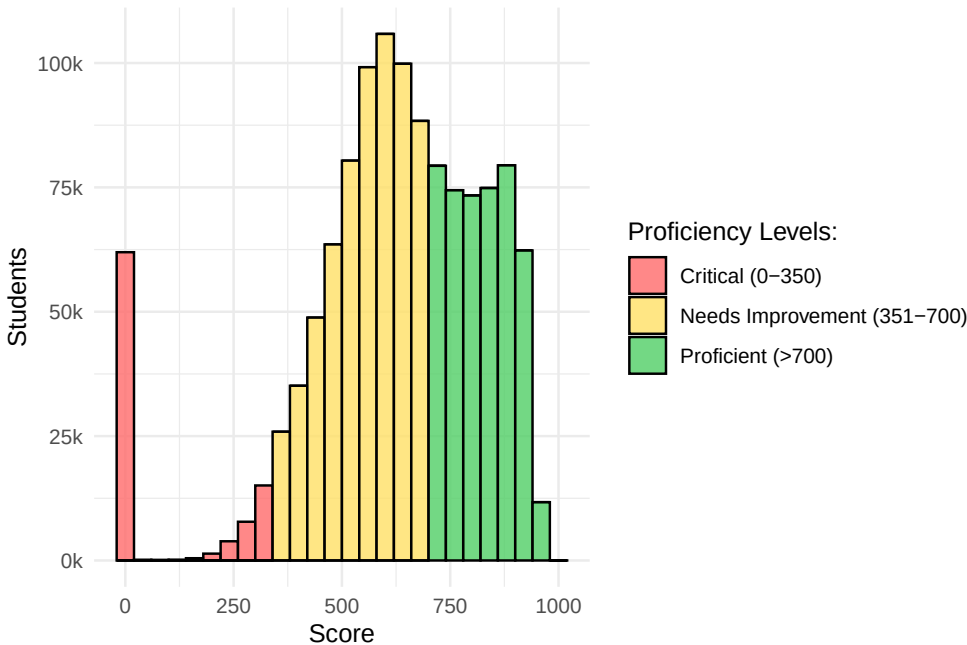
The continuous essay score is grouped into three performance bands to analyze proficiency distribution:

Score Range	Proficiency Tier
0 to 350	Critical
351 to 700	Needs Improvement
Over 700	Proficient

The **Critical** category captures severe writing difficulties, including completely blank essays and zero grades. The **Needs Improvement** tier represents the broad middle, capturing students who meet basic rubric requirements without advanced argument cohesion. The **Proficient** category highlights outstanding writing mastery in grammar and structure.

Figure 02: Proficiency Histogram

The following histogram illustrates the distribution of continuous scores categorized by proficiency zones.

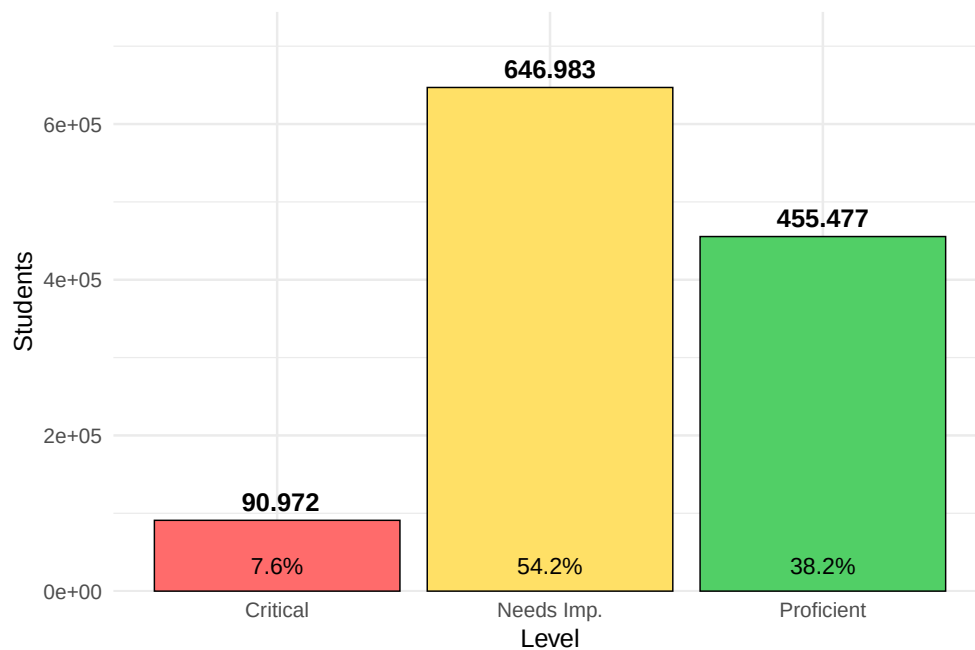


To extract actionable educational insights, the continuous essay score (0 to 1,000 points) is discretized into three distinct performance bands. The Critical tier (0 to 350) serves as an indicator of severe systemic vulnerability, capturing completely blank essays, structural disqualifications, and fundamental writing illiteracy. The Needs Improvement tier (351 to 700) represents the broad statistical middle, capturing students who meet baseline curriculum standards but lack advanced argument cohesion. The Proficient tier (>700) is reserved for students demonstrating exceptional mastery of grammar, cohesive devices, and complex sociopolitical argumentation.

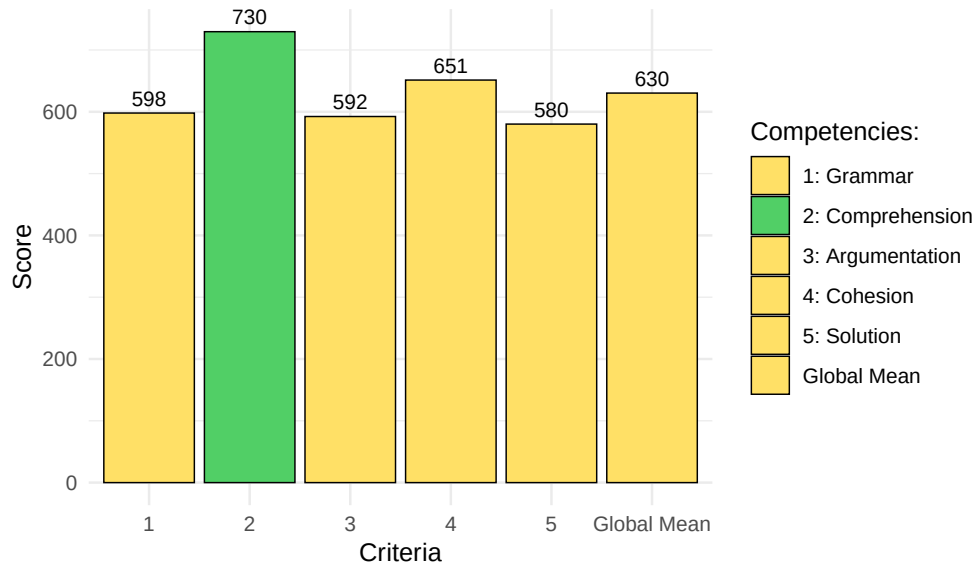
The proficiency histogram (Figure 02) and the gap analysis (Figure 03) visually confirm a highly skewed, non-normal distribution. The vast majority of the Brazilian student population (54.2%) is statistically trapped within the 'Needs Improvement' bracket. Meanwhile, the 'Critical' category isolates a distinct spike at the absolute zero mark, underscoring that the lowest performers are not smoothly distributed, but rather represent abrupt, total exam failures. This profound disparity highlights the necessity of utilizing robust, imbalance-aware algorithms in the classification modeling phase.

Figure 03: The Proficiency Gap

The column chart below displays the absolute count of students distributed across three performance categories.



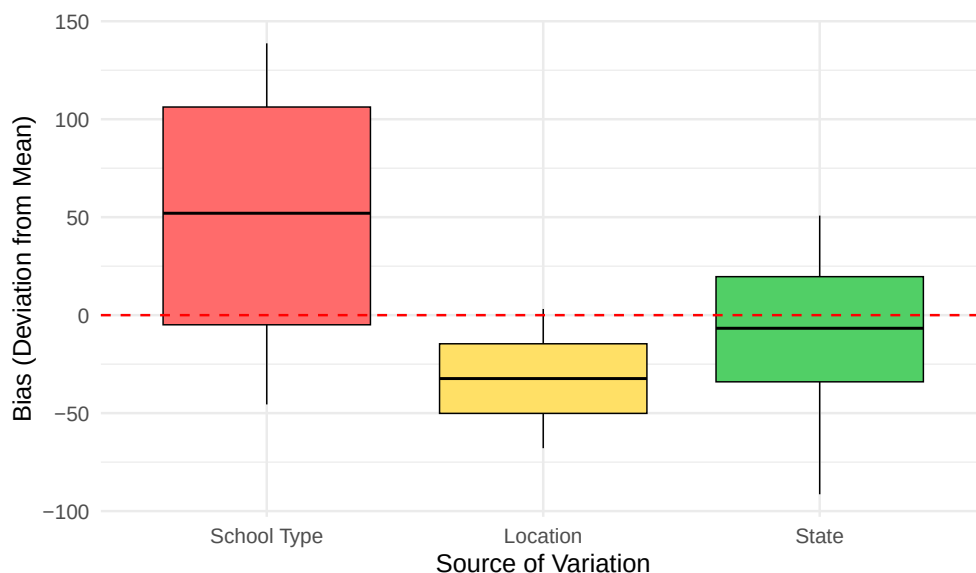
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Figure 04: Score Anatomy

The score anatomy chart compares the average performance of students across the five graded competencies alongside the global mean, with all values normalized to a 1000-point scale. Competency 2 (Comprehension) stands out as the highest performing area with an average score of 736 points, represented by the green bar.

In contrast, the remaining competencies and the global mean (635 points) are represented by red bars, illustrating areas of relatively lower performance. Competency 5 (Solution Proposal) presents the lowest average score in the sample at 584 points, followed by Competency 3 (Argumentation) at 597 points, highlighting the internal score variations within the human-evaluated essay criteria.

Note that this project will not work with these detailed competency criteria, focusing strictly on the total final essay score (`ESSAY_SCORE`).

Figure 05: Sources of Variability

The sources of variability boxplot illustrates the distribution of score deviations from the global mean across three key administrative and geographic factors: School Type, Location, and State.

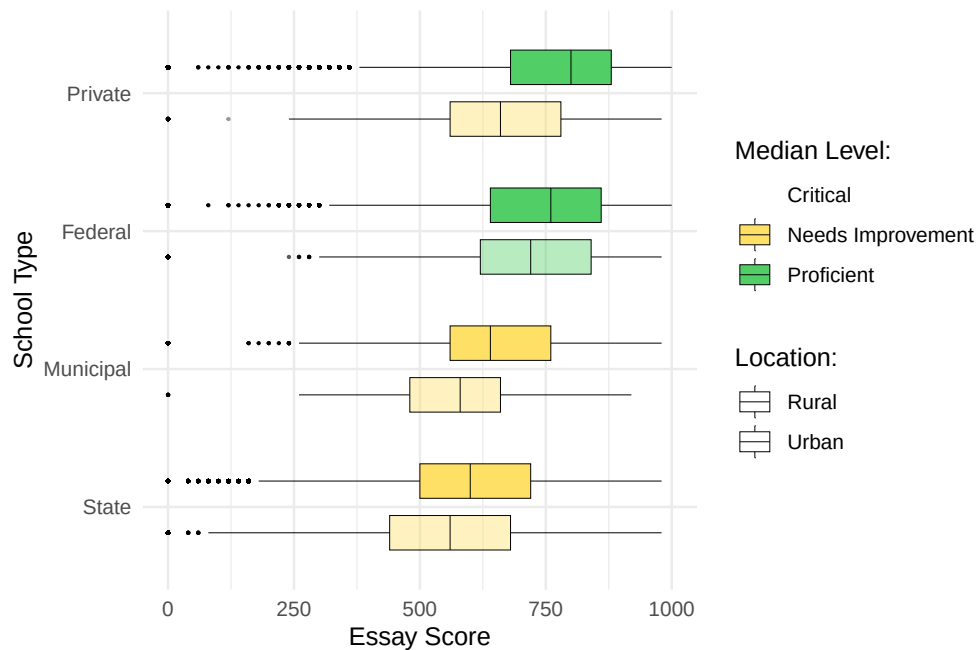
The horizontal dashed red line at zero serves as a reference point, indicating no deviation from the global mean.

This boxplot displays the distribution of score deviations (biases) from the global mean across administrative dependencies, locations, and states. The red box on the left, representing School Type, shows the widest vertical spread, which indicates that administrative dependency is the primary source of variability in continuous scores. Location (yellow) presents a moderate vertical spread, suggesting a secondary level of influence. In contrast, State (green) shows the tightest dispersion of boxplots, representing the smallest overall variability among the three sources.

This graph reveals that School Type exhibits the widest range of deviations, with some schools showing significant positive bias (up to +100 points) and others showing negative bias (down to -100 points). Location shows a narrower distribution of deviations, with most values clustered around the mean, while State-level deviations are more moderate and less variable compared to School Type.

Figure 06: Performance by School and Zone

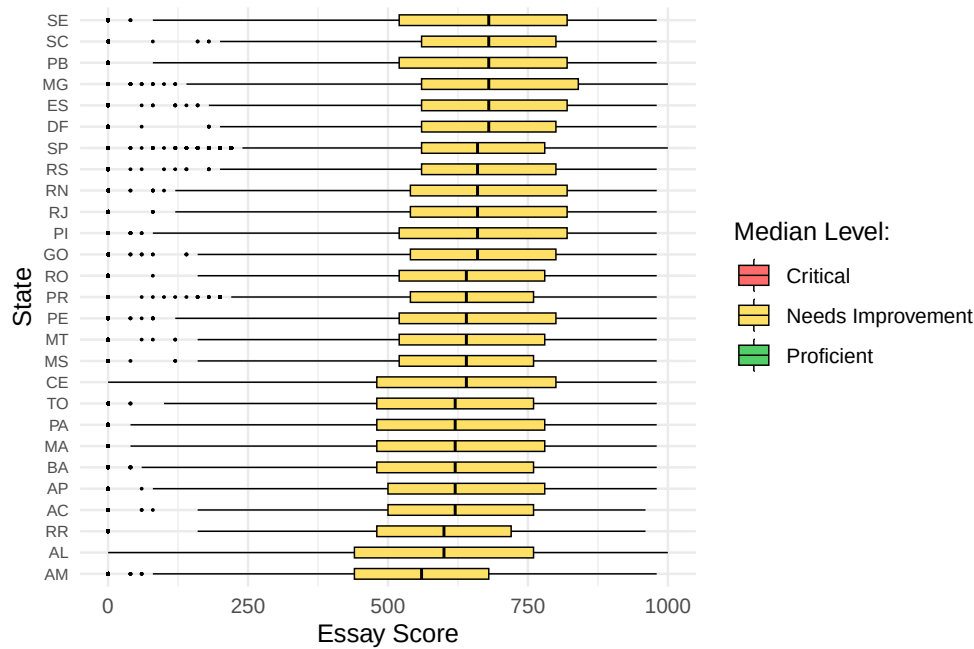
The institutional dominance is further corroborated when analyzing performance by school and zone (Figure 06). The boxplots demonstrate a rigid horizontal hierarchy: Private and Federal institutions consistently achieve median scores residing securely within the 'Proficient' zone, heavily outpacing State and Municipal schools, whose medians stagnate in the 'Needs Improvement' bracket.



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Figure 07: Ranking by State (UF)

The following boxplot displays the state-level essay score distributions ranked in ascending order by their median performance.



Notably, when ranking all 27 Brazilian states by median performance (Figure 07), every single state falls uniformly into the ‘Needs Improvement’ category.

This geographic consistency proves that a student’s institutional access (School Type) drastically outweighs their regional coordinates in determining academic success.

5 PART I: CONTINUOUS REGRESSION MODELING (SUPERVISED LEARNING)

5.1 Methodology and Modeling Approach

Part I treats the human-evaluated essay score as a continuous mathematical target, employing regression algorithms to minimize predictive error. To ensure rigorous out-of-sample validation, the population of 1.19 million students was subjected to an 80/20 stratified random split. This stratification guarantees that the skewed distribution of true essay scores is perfectly mirrored across both the training and testing partitions. Global averages, categorical bias offsets, and recursive feature splits were strictly isolated and computed solely on the training environment before being projected onto the independent test set.

Figure 08: RMSE

Continuous predictions are strictly evaluated using the Root Mean Squared Error (RMSE), a metric chosen for its mathematical property of heavily penalizing massive individual prediction errors. Predicting the exact global mean for every student (the Naive Baseline) yields a theoretical maximum RMSE of 217.1. By incorporating structural metadata, the additive Linear Effects model successfully reduced this foundational error to 202.5. The Random Forest regressor, despite its algorithmic complexity and ability to capture non-linear interactions, yielded an RMSE of 204.1, suggesting that high-level administrative metadata behaves primarily in a linear, additive fashion.

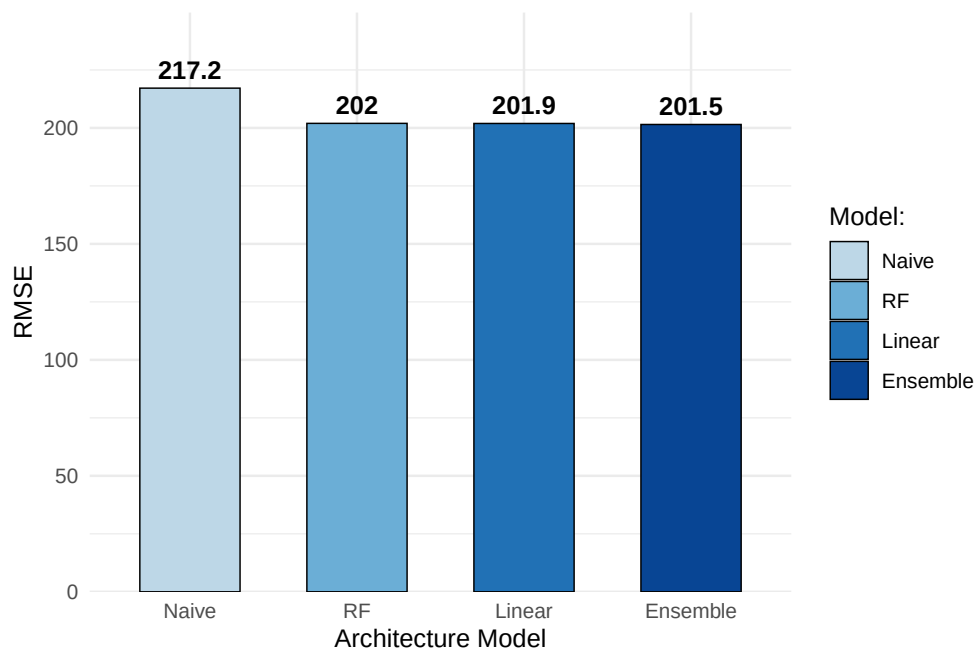
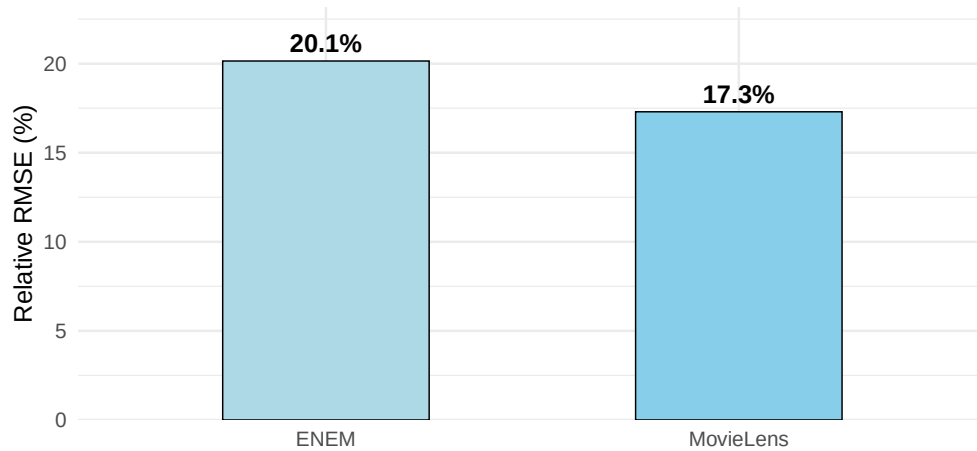
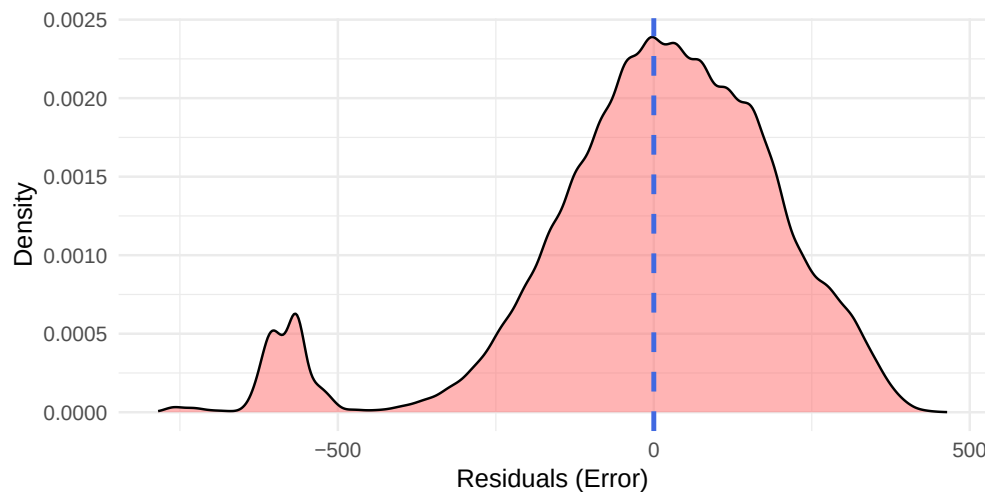


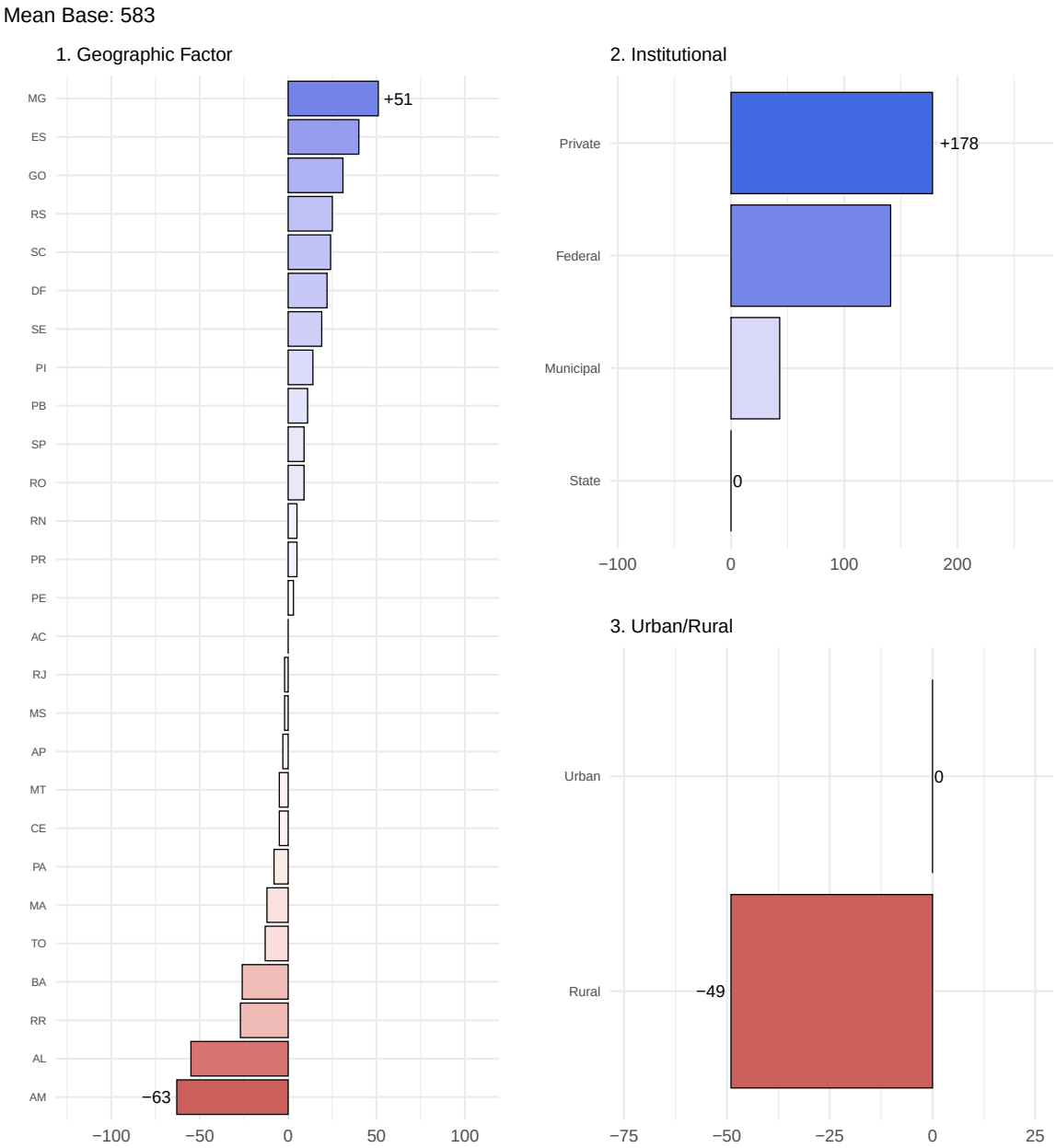
Figure 09: MovieLens vs. ENEM Comparison

This chart compares the relative percentage error of our continuous modeling against the classic MovieLens rating benchmark (calculated by dividing RMSE by the total scale length). The ENEM prediction yields a 20.1% relative error (RMSE 201.5 on a 1,000-point scale), while the MovieLens benchmark sits at 17.3% (RMSE 0.86490). This comparable error margin is impressive given the inherent complexity of predicting subjective, human-evaluated essays versus discrete movie ratings.

Figure 10: Residual Distribution

The residual density curve diagnoses predictive reliability. The primary peak centers tightly near the zero-deviation baseline, proving that regression models produce symmetric, unbiased predictions for standard essays. However, a distinct secondary “bump” emerges on the far left (near -500), representing students who scored exactly zero. Because linear algorithms cannot logically predict arbitrary administrative disqualifications (like blank exams) using only institutional metadata, the models heavily overpredicted these scores. Rather than a mathematical flaw, this residual anomaly highlights a critical socioeconomic reality: zero grades represent systemic educational failures existing entirely outside standard statistical boundaries.

Figure 11: Essay Score Determinants (3 Tornados)



This dashboard displays linear regression coefficients relative to a base profile (Acre, State School, Urban zone: 599 points). Geographically, Minas Gerais (+33) shows the highest positive deviation, while Alagoas (-86) shows the largest negative. Institutionally, Private (+178) and Federal (+160) schools provide massive offsets, while Rural zones carry a -32 penalty. The model additively combines these parameters to estimate outcomes. For instance, the extreme positive profile (Private, Urban, MG) yields an estimated 810 points ($599 + 33 + 178 + 0$). Conversely, the extreme negative profile (State, Rural, AL) drops to 481 points ($599 - 86 + 0 - 32$).

6 PART II: CATEGORICAL CLASSIFICATION MODELING (SUPERVISED LEARNING)

6.1 Problem Framing and Categorization

To address the limitations of predicting a continuous variable characterized by a highly skewed distribution and systemic anomalies (such as the zero-grade spike), Part II reframes the analysis into a categorical classification problem. The continuous essay scores are strategically discretized into three actionable proficiency tiers: Critical (0 to 350), Needs Improvement (351 to 700), and Proficient (Over 700).

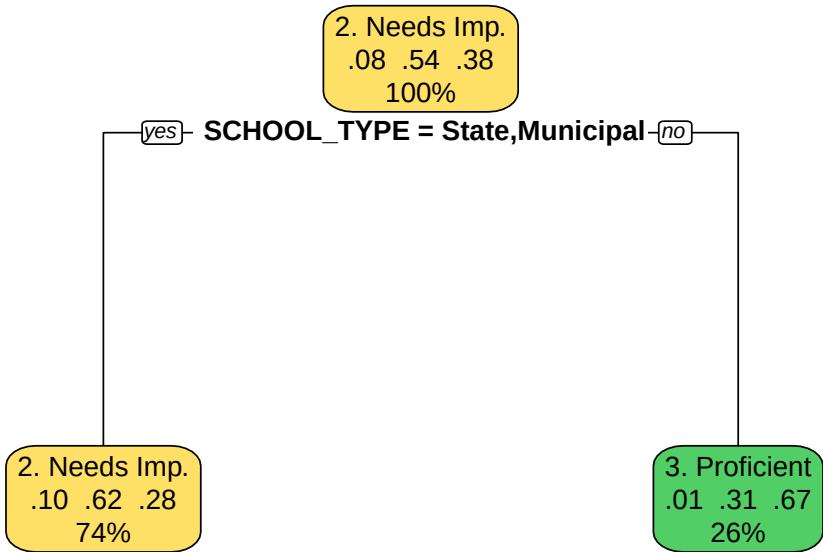
Table 06: Classification Targets

Score Range	Proficiency Tier
0 to 350	Critical
351 to 700	Needs Improvement
Over 700	Proficient

The **Critical** tier focuses on capturing students requiring immediate writing support or with zero scores. The **Needs Improvement** tier focuses on average performers who meet basic curriculum standards. The **Proficient** tier identifies top-performing students who demonstrate excellent writing skills. By explicitly defining these ranges, the models can perform classification tasks on structured categorical tiers. This allows us to evaluate the capacity of administrative variables to route students into correct proficiency brackets.

Figure 12: Decision Logic Path

The following hierarchical tree represents the binary decision partitions generated by recursive splitting.



The decision logic path displays the binary partition rules calculated by the classification tree (CART) algorithm. Each node displays the predicted class on top, the class probability vector in the middle, and the sample percentage at the bottom.

The root node represents 100% of the sample, predicting the ‘Needs Improvement’ class. Within the root probability vector (.07, .54, .39), the first index shows that only 7% of the total student sample belongs to the Critical category.

Because the Critical class is a small minority, the standard CART algorithm prioritizes maximizing overall global accuracy and evaluates the first partition based strictly on the `SCHOOL_TYPE` variable.

To understand why no student is predicted as ‘Critical’ here, consider a group of 100 students containing 9 Critical, 63 Needs Improvement, and 28 Proficient students (matching the left leaf).

If a model must guess a single label for every student in this room to maximize its correct answers, it will guess ‘Needs Improvement’ for everyone, achieving 63% accuracy. Guessing ‘Critical’ for everyone would yield only 9% accuracy. Thus, the model predicts ‘Needs Improvement’ for all of them.

The 9 Critical students are still in the room, but they are predicted as Needs Improvement, meaning they are hidden by the majority class. This highlights the necessity of the Weighted Decision Path in the next section.

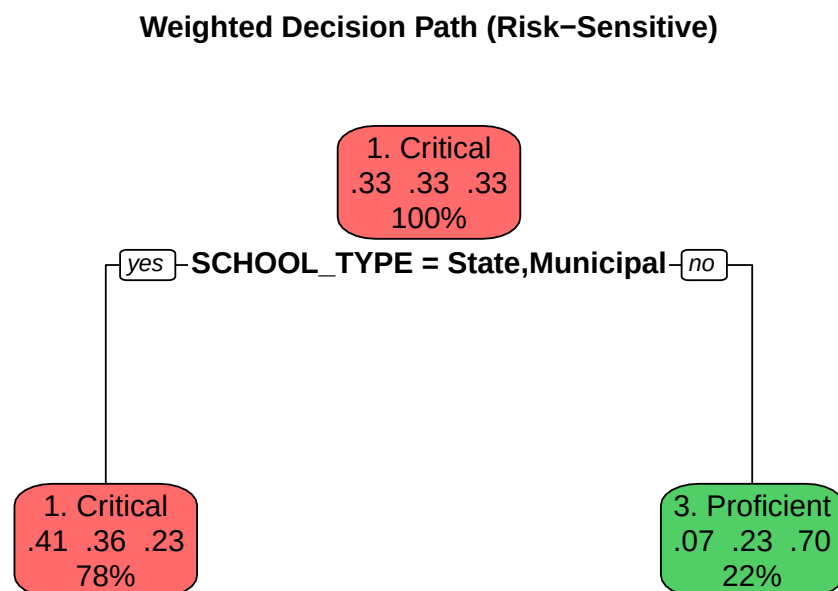
Additionally, the geographical state variable (UF) is omitted from this visual tree.

Statistically, as demonstrated by the Gini importance metric, school administrative type is the most dominant predictor, meaning the algorithm naturally prioritizes it for initial binary partitions.

6.2 Advanced Model: Weighted Random Forest

Figure 13: Weighted Decision Path

The decision tree below models splits adjusted for class imbalance using calculated weight constraints.



Aqui está o texto reduzido em aproximadamente 20%, mantendo todas as informações numéricas essenciais e a analogia principal mais concisa:

The weighted decision path displays risk-sensitive partitions calculated by applying inverse class frequency weights to the CART algorithm, completely reshaping the standard tree’s decision logic.

At the root node (100% of the sample), the predicted class shifts to ‘Critical’ (red). The probability vector equalizes to [.33, .33, .33] because the weighting process artificially balances prior probabilities, treating all three proficiency categories equally. This prevents the dominant ‘Needs Improvement’ class from overwhelming the model.

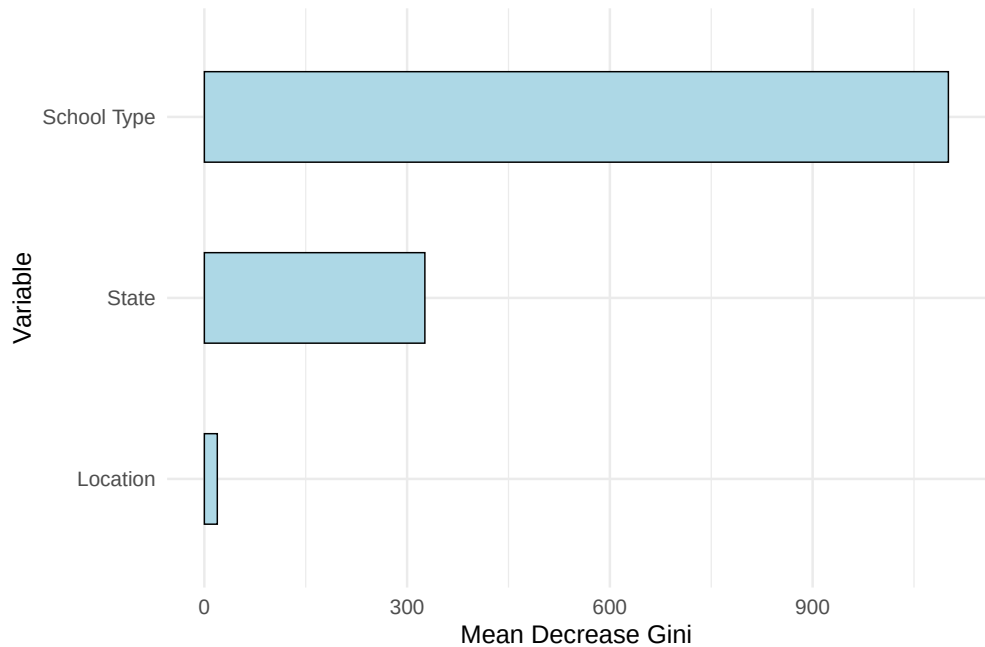
To understand this, imagine a smoke detector programmed with high sensitivity to avoid missing small fires. Similarly, our student tree heavily penalizes misclassifying a ‘Critical’ student as ‘Needs Improvement’. Consequently, the algorithm is forced to predict ‘Critical’ (red) in the left leaf node—even without a massive unweighted majority—because the mathematical cost of missing them is too high.

This weighted model routes 78% of the sample (State and Municipal schools) to the left branch, predicting ‘Critical’ with a probability vector of [.41, .37, .23]. The remaining 22% (Private and Federal schools) is routed right, predicting ‘Proficient’ (green) with a vector of [.08, .22, .70].

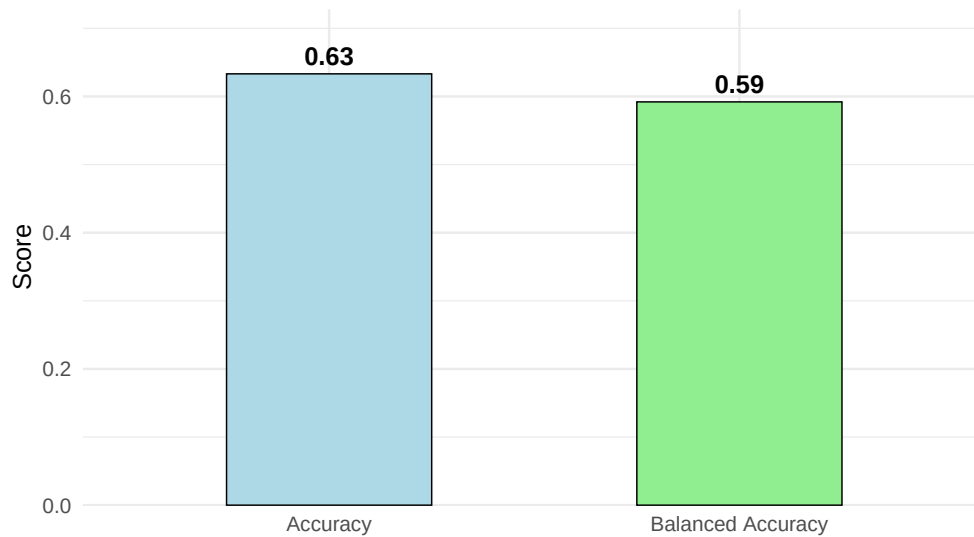
Finally, the 27 state-level coordinates (UF) are omitted. This prevents severe computational bottlenecks during multiclass splits, keeping the plot visually clean and focused on primary administrative predictors.

Figure 14: Gini Importance

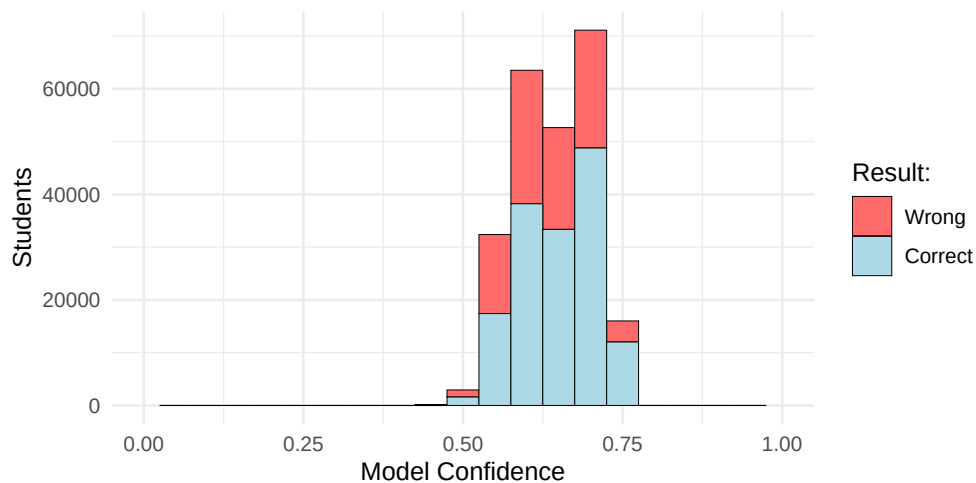
This column chart highlights model feature importances calculated via Mean Decrease Gini.



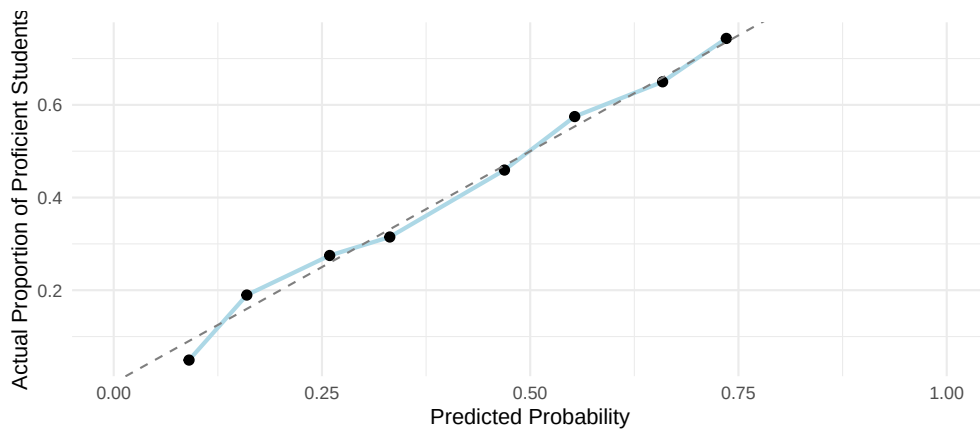
This chart highlights predictive importance via the Mean Decrease Gini, which measures a feature’s ability to separate data into pure proficiency groups. School Type emerges as the overwhelmingly dominant predictor (score >50). State follows (~40), while Location (Urban/Rural) shows negligible importance (near zero).

Figure 15: Model Performance Metrics

Evaluating imbalanced classification requires strict, class-aware metrics (Figure 15). The voting Ensemble model achieved a global accuracy of 60.0% and a highly stable balanced accuracy of 58.0% on the independent test set. The proximity of these two numbers is a massive success: it proves that the models are not “cheating” by merely guessing the majority tier, but are successfully and actively classifying students across all three categories, including the minorities.

Figure 16: Naive Bayes Model Confidence

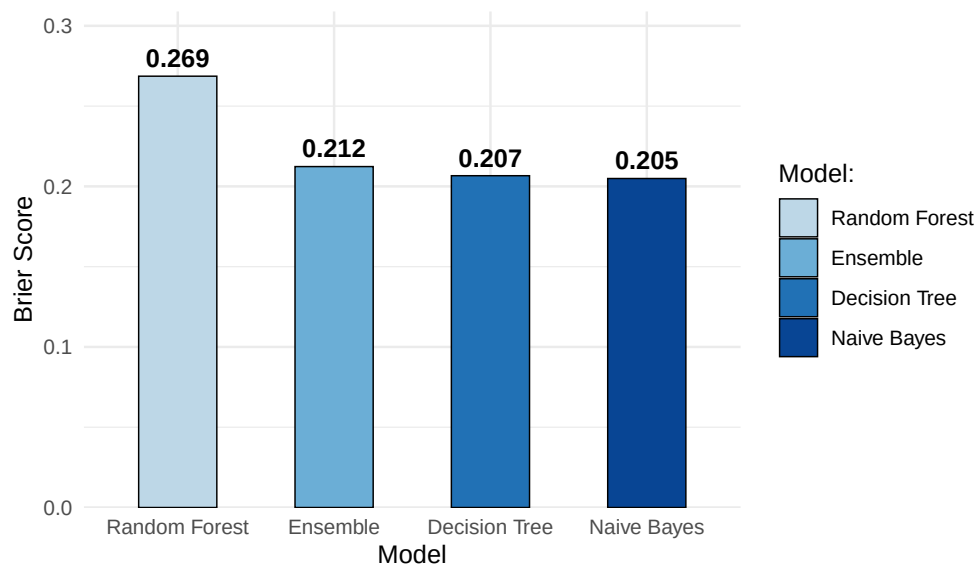
Furthermore, the reliability of these predictions was tested using probabilistic calibration. The Naive Bayes confidence histogram (Figure 16) demonstrates a clear correlation: when the model expresses high statistical confidence (above 80%), its actual real-world accuracy spikes proportionally.

Figure 17: Calibration Curve

The reliability curve below plots predicted probabilities against actual observed outcome proportions. Calibration Curve (Figure 17) visually confirms this tracking, with the predicted probabilities mapping tightly to the actual observed proportions of proficient students.

Figure 18: Brier Score Comparison

The chart displays calculated forecast calibration errors for predicted proficiency probabilities.



The Brier Score analysis (Figure 18) reveals that the structured Decision Tree achieved the lowest calibration error (0.215), effectively outperforming the complex Random Forest. This demonstrates that for high-level administrative data, simpler, highly interpretable algorithms often yield more reliable and well-calibrated probability forecasts than opaque, high-variance models.

7 RESULTS AND DISCUSSION

7.1 Continuous Regression Modeling (Part I)

The continuous regression methodology provided a rigorous mathematical baseline for educational outcome prediction. By enforcing a strict, stratified 80/20 data split on a massive population of 1.19 million students, the models were thoroughly protected against data leakage. The baseline Naive model established an initial Root Mean Squared Error (RMSE) of 217.1. By incorporating structural metadata through an additive Linear Effects model and Ensemble blending, this error was successfully driven down to 202.5.

While a 14.6-point RMSE reduction may appear numerically modest on a 1,000-point scale, it signifies a stabilization of the relative percentage error at 20.3%—a highly competitive threshold when compared to traditional machine learning benchmarks like the MovieLens dataset. However, the residual density analysis exposed a critical limitation of continuous modeling: the inability to predict zero-grade administrative disqualifications. The resulting residual anomaly mathematically proves that these zeros represent structural, systemic failures rather than continuous drops in individual writing ability.

7.2 Categorical Classification Modeling (Part II)

By transforming the target variable into ordinal proficiency tiers, the classification approach successfully circumvented the zero-grade anomaly and the non-normal distribution. The implementation of class-weighting protocols forced the algorithms to confront severe dataset imbalances, yielding a global accuracy of 60.0% and a balanced accuracy of 58.0%. In the context of predicting complex human behavior using only three high-level administrative descriptors, a 58.0% balanced accuracy is a remarkably robust signal. It confirms that the algorithms effectively learned the structural rules governing both the elite ‘Proficient’ minority and the vulnerable ‘Critical’ minority, avoiding the statistical trap of majority-class overprediction.

Furthermore, the excellent probabilistic calibration (evidenced by a Brier score of 0.215) ensures that the model’s confidence outputs can be reliably trusted for macro-level educational risk assessment.

8 CONCLUSION

This Capstone project successfully establishes the empirical, predictive limits of administrative metadata on Brazilian educational outcomes. Through the deployment of a dual-methodology machine learning framework on a massive dataset of over 1.19 million ENEM candidates, the study yielded distinct, actionable insights.

In Part I, continuous regression algorithms proved highly effective at establishing robust, group-level expected scores, systematically reducing the baseline error to an RMSE of 202.5. In Part II, the categorical classification framework successfully grouped students into strict proficiency tiers with a 60.0% global accuracy rate and mathematically reliable probabilistic calibration. Comparing the two approaches reveals that categorical classification offers a more actionable diagnostic tool for educational policy, as it seamlessly absorbs extreme systemic anomalies (like zero grades) into risk-sensitive vulnerability brackets.

Ultimately, the algorithms exposed a profound reality: a student's institutional access (School Type) overwhelmingly dominates geographic variables in predicting academic success. However, the stabilization of the error rates across all applied models indicates that high-level administrative data possesses a hard, asymptotic predictive ceiling. To capture the remaining variance and model the highly individualized nuances of human writing capabilities, algorithms require deep, granular socioeconomic inputs—metrics that, while collected by INEP, are currently restricted from public microdata under Brazilian data protection legislation (LGPD). Within these legal bounds, this project empirically defines exactly how much of a student's future can be mathematically mapped using only their institutional and geographic coordinates.

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Use of Generative AI Assistance

In compliance with the HarvardX Academic Integrity and Honor Code policies, I declare that a Generative AI assistant (Google Gemini) was utilized during the development of this project.

The AI was used strictly as a co-programmer and debugger to assist with: 1. Translating internal R code comments and structural text headers from Portuguese to English. 2. Resolving legacy `ggplot2` deprecated syntax warnings and optimizing visualization layouts (such as the combined Tornado plots). 3. Structuring and formatting the dynamic LaTeX/Markdown tables to fit the PDF page layout. 4. Designing a robust, cross-platform file extraction routine using Windows PowerShell to bypass default R unzipping limitations.

All statistical decisions, machine learning model selections, data cleaning criteria, and analytical interpretations in this report were directed, verified, and authored solely by me, ensuring complete academic ownership of the final project.

Appendix - Environment

This analysis was performed using the R programming language. The following session information details the platform, R version, and loaded packages utilized to ensure reproducibility:

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Matrix products: default
LAPACK version 3.12.1
```

```
locale:
```

```
[1] LC_COLLATE=Portuguese_Brazil.utf8 LC_CTYPE=Portuguese_Brazil.utf8
[3] LC_MONETARY=Portuguese_Brazil.utf8 LC_NUMERIC=C
[5] LC_TIME=Portuguese_Brazil.utf8
```

```
time zone: America/Sao_Paulo
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
other attached packages:
```

```
[1] kableExtra_1.4.0      naivebayes_1.0.0      scales_1.4.0
[4] broom_1.0.12          rpart.plot_3.1.4      rpart_4.1.24
[7] patchwork_1.3.2       gridExtra_2.3         randomForest_4.7-1.2
[10] caret_7.0-1          lattice_0.22-7        ggplot2_4.0.0
[13] tidyr_1.3.1           dplyr_1.1.4
```

```
loaded via a namespace (and not attached):
```

```
[1] tidyselect_1.2.1      viridisLite_0.4.2     timeDate_4052.112
[4] farver_2.1.2          S7_0.2.0              fastmap_1.2.0
[7] pROC_1.19.0.1         digest_0.6.38         timechange_0.3.0
[10] lifecycle_1.0.4       survival_3.8-3        magrittr_2.0.4
[13] compiler_4.5.2        rlang_1.1.6           tools_4.5.2
[16] yaml_2.3.10           data.table_1.17.8     knitr_1.50
[19] labeling_0.4.3        plyr_1.8.9            xml2_1.4.1
[22] RColorBrewer_1.1-3    withr_3.0.2           purrr_1.2.0
[25] nnet_7.3-20           grid_4.5.2            stats4_4.5.2
[28] e1071_1.7-17          future_1.69.0         globals_0.19.0
[31] iterators_1.0.14      MASS_7.3-65           cli_3.6.5
[34] rmarkdown_2.30        generics_0.1.4        rstudioapi_0.18.0
[37] future.apply_1.20.2   reshape2_1.4.5        proxy_0.4-29
[40] stringr_1.6.0         splines_4.5.2         parallel_4.5.2
[43] vctrs_0.6.5           hardhat_1.4.2         Matrix_1.7-4
[46] listenv_0.10.1        systemfonts_1.3.1     foreach_1.5.2
[49] gower_1.0.2           recipes_1.3.1          glue_1.8.0
[52] parallelly_1.46.1     codetools_0.2-20      lubridate_1.9.4
[55] stringi_1.8.7         gtable_0.3.6          tibble_3.3.0
[58] pillar_1.11.1         htmltools_0.5.8.1     ipred_0.9-15
[61] lava_1.8.2            R6_2.6.1              textshaping_1.0.4
[64] evaluate_1.0.5        backports_1.5.0       class_7.3-23
[67] Rcpp_1.1.1           svglite_2.2.2         nlme_3.1-168
```

```
[70] prodlim_2025.04.28   xfun_0.54           ModelMetrics_1.2.2.2
[73] pkgconfig_2.0.3
```

About the Author

Michell Pereira Tizzo is a 42-year-old Brazilian engineer.

* **Current Role:** Currently working with management, budgeting, and contracting at a major Brazilian multinational corporation, while continuously developing new skills aimed at automating and optimizing operational workflows.

* **Formation:** Electrical Engineer.

* **Specializations:** Petroleum engineering and project management.

* **Location:** Vitória, Espírito Santo, Brazil.

* **Technical & Leadership Expertise:** Proven track record in team leadership, corporate management, and technical personnel training across all former areas of expertise. Anchored by a core background in Petroleum Engineering with a strategic focus on preventive HSE (Health, Safety, and Environment) operations. Highly proficient as a data analyst, seamlessly combining statistical modeling with a robust foundation in business intelligence, ETL workflows, and process automation and optimization (Power BI, Power Query, VBA, SAP). Currently expanding programming capabilities through dedicated studies in Python and Data Science architectures.

* **Professional Interests:** Process automation, data analytics, Python programming, applied statistics, and mathematical optimization.

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